Simultaneous Acquisition and Sensing of ECG and IPG Signals for measurement of Pulse Transit Time

Group MON-04 — Milestone 3

Yashvardhan, Krunal, Sudhindra, Vinay, Jahnvi

March 21, 2026

IIT Bombay

Feedback from Milestone-2

Feedback	Action taken
Switch from a 1S Li-ion power source to 2S	Since most the components can take 5V input and it would be easier to step down 2S to 5V, we have changed our battery and protection circuit from 1S to 2S
Maximize copper area on the PCB for noise filtering and heat dissipation	We maximized the copper area by using ground planes on both layers of our two-layered PCB connected at multiple locations with via stitchings.
Use wired channel to send waveforms to laptop	We have used a wired UART-to-USB connection to stream real-time data to the laptop; minimizes power usage and eliminates the latency common in wireless modules.
Software for bio-signal processing should be main focus of our project	Since the MAX30001 handles the raw signal acquisition, our primary contribution lies in processing that data to derive meaningful medical parameters.

Video Link for Proof of Concept

Demonstration Video:

https://drive.google.com/drive/folders/1t-SZbwQfwB-TrOjXBIuESoD1K13y0rSp?usp=sharing

(Click the link to view the working proof of concept)

Update on the PCB

- **Progress level**: Submitted for fabrication at IIT PCB Lab
- **Dimensions**: 12.2 cm X 6.6 cm (Rectangular)
- **Key Learnings**:
  1. **Design for Manufacturability (DFM)**: The layout has to be made under the constraints of the manufacturer's technical limits, such as minimum track width and spacing, to ensure the board can be fabricated without any unwanted defects.
  2. **Thermal Management:** Larger copper areas help spread and dissipate heat from components like linear voltage regulators. On a 2-layer PCB, it is best to include ground planes on both layers and connect them with stitching vias to improve heat dissipation.
  3. **Component and Footprint Verification:** Before assigning footprints to schematic symbols, we must confirm that the components are available and double-check that the physical dimensions, pad arrangements, and SMD package sizes match our design exactly.

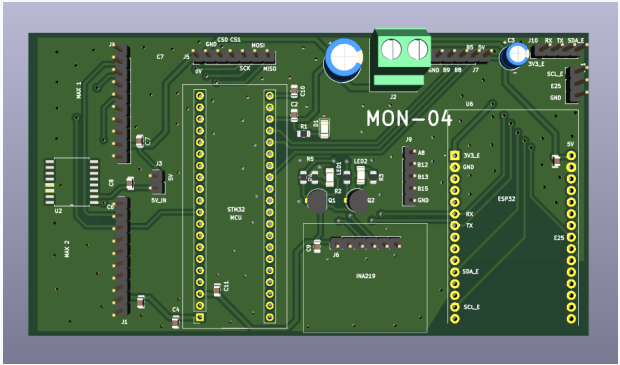
Update on the PCB

Design Choices & Justification:

1. **Two-layered PCB** with SMD components on top layer and Ground plane on Top/Bottom layer: Using ground planes on both sides helps to shield signals from noise and provide a stable return path. By placing all components on the top layer, the assembly processes for both SMD and Through-Hole components get simplified. Also having a continuous ground plane in the bottom layer dissipates heat away due to larger copper area.
2. **Modular PCB design:** Since the main goal of our project is building the software stack for bio-signal processing, the main objective of our PCB is to simply act as an integration unit for power and signal connections between different modules. This modular setup lets us focus on the algorithms and makes it easy to replace parts if they get damaged.

Update on the PCB

3. **Indicator LEDs & breakout pins for Debugging**: These serve as built-in test points, making it easier to monitor power & signal lines and troubleshoot issues with an oscilloscope or multimeter.
4. **Power stability**: We placed decoupling capacitors at the power inputs and outputs of each module to filter out electrical noise and keep the voltage steady.



CAD Prototype

- The enclosure is designed to integrate all system components, including a dedicated battery compartment and internal PCB mounting structure
- User interface elements include a toggle switch for power control, push buttons for mode selection, and a compact display for real-time status indication
- External interfacing is enabled through ports for power input, code upload, and electrode connections, along with organized slots for electrode storage
- Currently we are in the preliminary stage, focusing on dimension validation, component fitting, and orientation optimization before final fabrication

CAD Prototype: Design Visualization

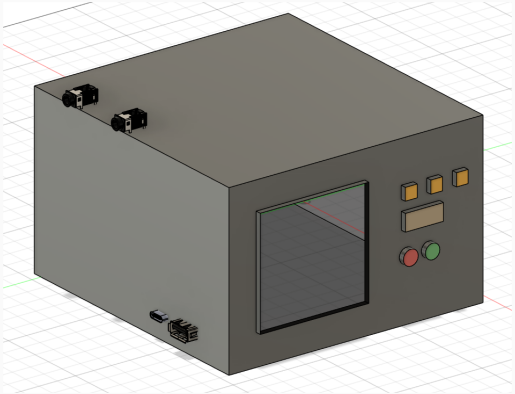


Figure 1: Outer Enclosure box with interactive buttons, display and ports

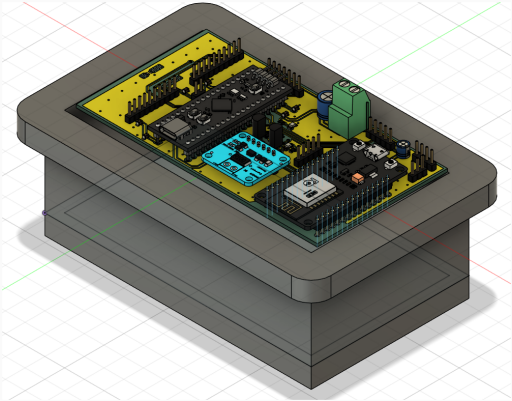
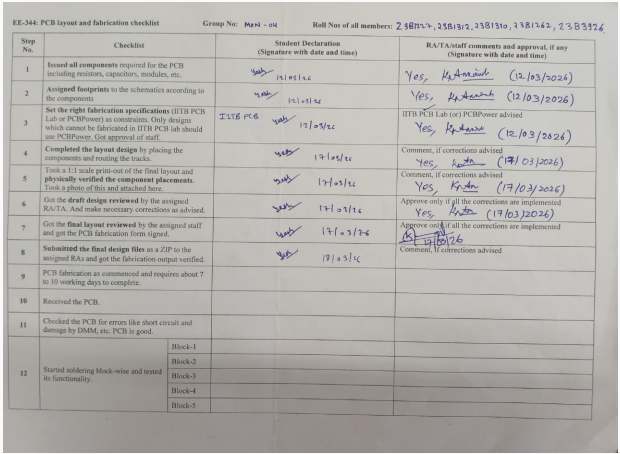


Figure 2: Internal PCB and battery holder

PCB layout and fabrication checklist

We are currently at 8th checkpoint.



EE-344: PCB layout and fabrication checklist Group No: MAN - 04 Roll Nos of all members: 23B1327, 23B1312, 23B1310, 23B1362, 23B3326

Step No.	Checklist		Student Declaration (Signature with date and time)	RA/TA/staff comments and approval, if any (Signature with date and time)
1	**Issued all components** required for the PCB including resistors, capacitors, modules, etc.		17/03/26	Yes, (12/03/2026)
2	**Assigned footprints** to the schematics according to the components		17/03/26	Yes, (12/03/2026)
3	**Set the right fabrication specifications** (IITB PCB Lab or PCBPower) as constraints. Only designs which cannot be fabricated in IITB PCB lab should use PCBPower. Got approval of staff.	IITB PCB	17/03/26	IITB PCB Lab (or) PCBPower advised. Yes, (12/03/2026)
4	**Completed the layout design** by placing the components and routing the tracks.		17/03/26	Comment, if corrections advised Yes, (17/03/2026)
5	Took a 1:1 scale print-out of the final layout and **physically verified the component placements**. Took a photo of this and attached here.		17/03/26	Comment, if corrections advised Yes, (17/03/2026)
6	Got the **draft design reviewed** by the assigned RA/TA. And make necessary corrections as advised.		17/03/26	Approve only if all the corrections are implemented Yes, (17/03/2026)
7	Got the **final layout reviewed** by the assigned staff and got the PCB fabrication form signed.		17/03/26	Approve only if all the corrections are implemented 17/03/26
8	**Submitted the final design files** as a ZIP to the assigned RAs and got the fabrication output verified.		17/03/26	Comment, if corrections advised
9	PCB fabrication as commenced and requires about 7 to 10 working days to complete.			
10	Received the PCB.			
11	Checked the PCB for errors like short circuit and damage by DMM, etc. PCB is good.			
12	Started soldering block-wise and tested its functionality.	Block-1		
		Block-2		
		Block-3		
		Block-4		
		Block-5		

Progress Made Against the Test Plan

1. PCB design phase completed with iterative improvements based on verification feedback
2. Signal processing, communication, and subsystem-level implementations are progressing in parallel as per plan
3. Power architecture refined to improve voltage stability and overall system reliability
4. Majority of modules validated individually; moving towards integrated system testing

Deviations and Justification:

- **PCB layout delayed by 1 week** due to critical feedback during verification:
  - Addition of ground planes on both sides of PCB
  - Removal of ground below ESP32 to avoid interference
- **Testing phase delayed** due to enhancement in power design:
  - Added LDO (AMS1117) after buck converter (LM2596S) to reduce voltage ripple
  - Updated plan: Step-down from 2S battery to 5.5V using buck converter, followed by LDO regulation to stable 5V

Work Distribution Until Final Demo

Team Member	Tasks and Contributions
Krunal	Signal processing: extraction of PTT and R-R peak detection from dual IPG waveforms; ensuring accuracy of physiological feature computation
Vinay	Communication system: enabling UART-to-USB wired channel for plotting waveforms; python code for user interface on the laptop; SPI protocol verification and debugging on PCB
Jahnvi	Mechanical and system integration: 3D CAD model validation (dimensions, strength, fit); foot detection algorithm; programming STM32 and implementation of RTOS
Sudhindra	Hardware implementation: PCB soldering and assembly; I2C verification for INA219 module; integration into final enclosure
Yashvardhan	System validation: BMS testing, power integrity checks, and end-to-end functional verification of the PCB and complete system

Plan ahead for Final Demo

1. After the PCB comes from manufacturing, would test the board for full functionality, fix any errors, and replace our breadboard prototype with the final PCB.
2. Would give the CAD model of the device casing for 3D-printing and verify its dimensions, strength, and fit before assembling the entire system using the CAD as the base.
3. We will run the software directly on live bio-signals as they are captured by the PCB in real-time.
4. After processing the bio-signals and using them to extract features such as the PTT (Pulse Transit Time), we relate this to medically useful health parameters.
5. We would compare our results with standard medical equipment to prove that our portable system provides accurate health readings. The final aim is to demonstrate that our simple and small electrical system can acquire useful health parameters.

Hardware Flow: Transition to STM32

Migrating from Arduino to STM32 Nucleo H7A3ZI

To achieve standalone, real-time Pulse Transit Time (PTT) calculations, we outgrew the 8-bit Arduino ecosystem. We migrated to the STM32H7 series to handle rigorous on-board signal processing:

- The inclusion of a Double-Precision Floating-Point Unit (FPU) allows us to execute math algorithms on raw data natively, without cycle penalties.
- Leveraging ARM's optimized DSP library to execute fast matrix operations and filtering directly on the microcontroller.
- Direct Memory Access offloads the CPU from SPI polling, ensuring zero-latency data acquisition from the dual AFEs while the core focuses entirely on algorithm execution.

Dual Chip Interfacing Strategy

Synchronized Data Acquisition via SPI

While previous milestones covered the internal register setup of the MAX30001, our current focus is managing the dual-data streams flawlessly on the STM32.

- **Bus Architecture:** Both AFEs share the main SPI bus but utilize independent GPIO Chip Selects (CS) managed by the STM32.
- **Interrupt-Driven Pipeline:** We utilize the hardware `INTB` pins from both AFEs. The STM32 RTOS uses these interrupts to trigger DMA read sequences the exact microsecond a new sample enters the AFE FIFO.
- **Timebase Alignment:** We periodically trigger a simultaneous `REG_SYNCH` command across both boards to zero-out internal timers, ensuring the ECG and IPG waves are mapped to an identical software timebase for strict PTT accuracy.

On-Board Signal Processing (DSP)

Shifting the Computational Load to the MCU

Previously, we relied on a Python GUI to clean the raw serial stream. We have now moved this filtering pipeline entirely onto the STM32 to create a standalone device.

- **Bypassing Hardware Limits:** We bypass the aggressive internal hardware filters of the MAX30001 to capture maximum raw bandwidth, feeding the raw ADC values directly into the STM32's memory.
- **Phase-Linear FIR Implementation:** We implemented a custom Finite Impulse Response (FIR) Moving Average.
- **Why Phase Linearity Matters:** Unlike standard IIR filters, the FIR design ensures all frequency components are delayed equally. This prevents artificial shifting of the waveform foot, which would otherwise introduce massive error dispersion into our Δt calculation.

Matlab Simulations: R-Peak Detection

To calculate an accurate Pulse Transit Time, we must precisely identify R-peak in the ECG stream. We implemented a modified Pan-Tompkins DSP algorithm:

1. Eliminates baseline wander and high-frequency muscle noise while strictly preserving the QRS phase timing.
2. Highlights the steep slopes of the R-wave and non-linearly amplifies the signal to ensure all data points are positive.
3. Aggregates the QRS energy into a distinct, measurable curve (using a 150 ms window).
4. Applies statistical thresholding to locate the integrated peak, then searches the raw buffer to tag the exact original R-peak index.

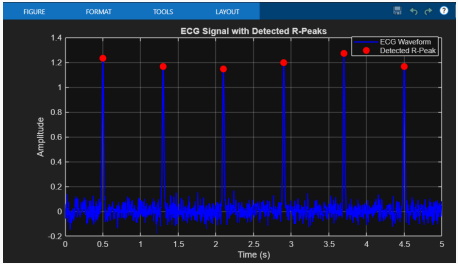


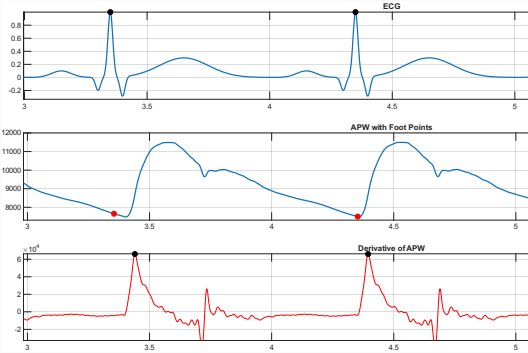
Figure 3: Pan-Tompkins Pipeline Validation

MATLAB Simulations: Algorithm Validation

Prototyping the IPG Foot Detection Offline

Before porting the complex detection math to the STM32, we simulated our foot detection approach in MATLAB using the datasets acquired from our new on-board filters.

- **Algorithm Testing:** Evaluated both the *Second Derivative Method* and the *Cubic Polynomial Fit* method for identifying the exact start of the Arterial Pulse Wave (APW).
- **Search Window Logic:** The simulation uses the detected ECG R-peak to establish a bounded search window (300ms) for the subsequent BioZ foot.
- **Result:** The polynomial fit provided the highest resistance to noise artifacts, confirming our mathematical approach before C-code



Foot detection under various SNR conditions

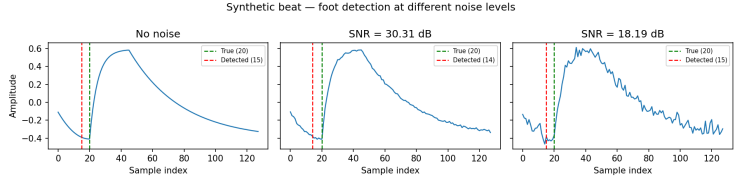
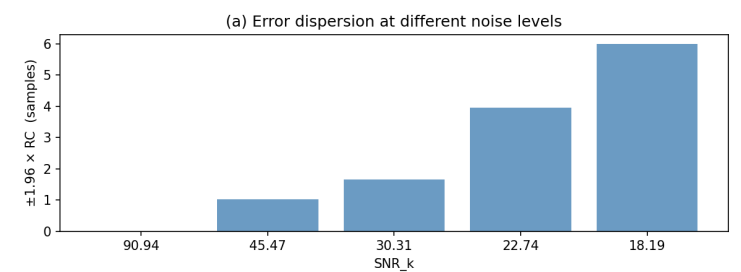


Figure 4: Synthetic beats



Next Steps: Blood Pressure Estimation, Reference Paper

Cuffless BP Estimation using PTT/PAT

With synchronized acquisition of ECG, APW (arterial pusle wave signals, the next step is to estimate Blood Pressure using Pulse Transit Time (PTT) and Pulse Arrival Time (PAT).

- **Physiological Basis:** Blood flow in arteries can be modeled as wave propagation in elastic tubes. BP is related to arterial stiffness, which affects Pulse Wave Velocity (PWV) and hence PTT.
- **Nonlinear BP–PTT Model:** Instead of a linear approximation, BP is modeled as:

$$\mathrm{BP} = a_0 + \sqrt{a_1 + \frac{a_2}{\mathrm{PTT}^2}}$$

  capturing the true nonlinear relationship between BP and PTT.
- **Estimation Pipeline:**
  1. Extract PTT or PAT (ECG–PPG)
  2. Segment signals and compute averaged values per interval
  3. Fit subject-specific model parameters using calibration data

Before Filtering

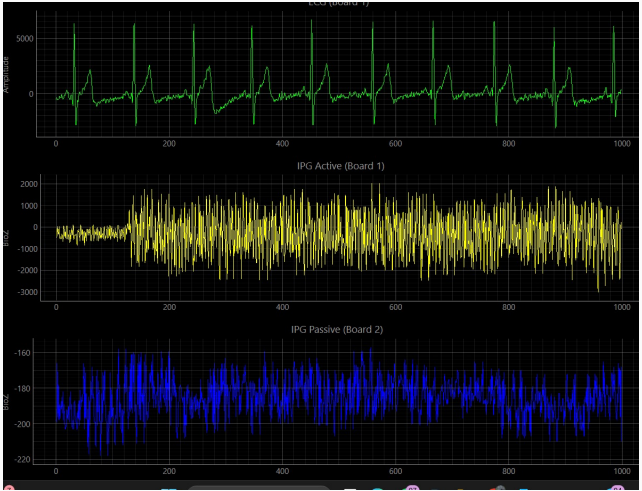


Figure 6: ECG AND BIOZ WAVERFORMS

After Filetring

Figure 7: ECG AND BIOZ WAVERFORMS